

Humidification

Pre-Lab Plan
Unit Operations Lab 1
01/20/2020

Group 2, Tuesday, PM section

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Commented [DL1]: 28.5/35

1. Experimental Objective:

Humidification is a simple process that was industrialized by chemical engineers to help them analyze experimental data in order to design cooling water and drying processes. In this experiment we will find the coefficients for heat and mass transfer and then plot it as a function of Reynolds number and try to compare it to an already existing correlation.

2. Experimental:

2.1 Apparatus:

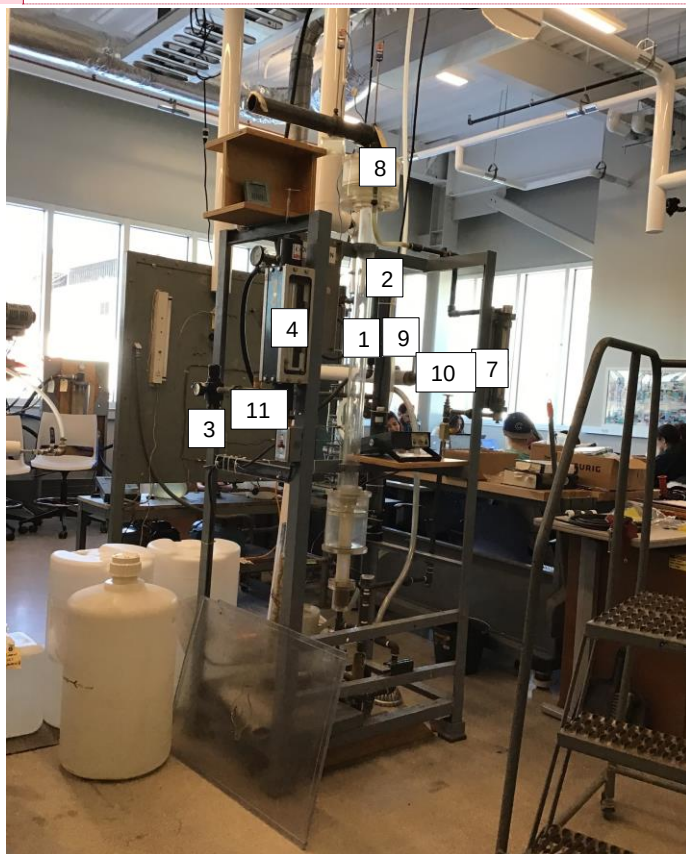


Figure 1: The important equipment of the experiment

Commented [DL2]: 3/5

The objective of this experiment is to study the transport of both mass and heat in a wetted wall column and obtain experimental values for the heat and mass transfer coefficients.

These values will be obtained through experimental methods and will be compared to values calculated using the theoretical equations.

The system variables that will serve as the independent variables for this experiment include liquid flowrate, air flowrate, and temperature of the incoming air.

Commented [AZ3]: Humidification is used as an industrial process itself, not just to analyze data.

Commented [DL4]: Use third-person statement.

Commented [DL5]: 3.5/5

A brief introduction of the apparatus is needed. You may include the flowsheet of the process.

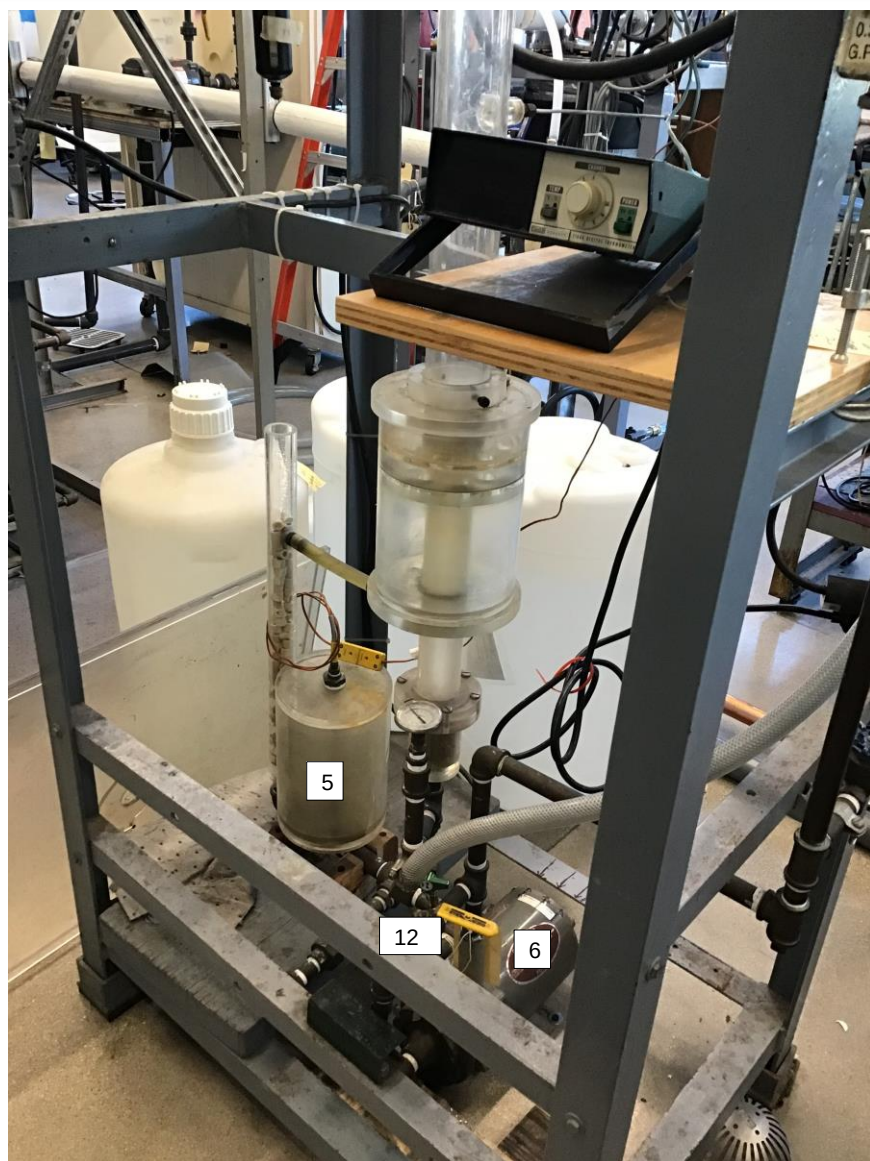


Figure 2: The important equipment of the experiment

Table 1: The important components used in the experiment

#	The important components	Description
1	The wetted-wall humidification column	2-inch diameter Plexiglas column 4 feet in length connected to an air flow and water feed system
2	House Supply	Supplies the system with air
3	Pressure regulator	Manipulates the pressure of air
4	Air rotameter	Measures the flowrate of air
5	Tank	Source of water supplied to the system
6	Centrifugal pump	Pumps the water from the holding tank into the system
7	Water rotameter	Measures the flowrate of water
8	Weir	Distributes the water to the side of the top of the column
9	Heat Exchanger	Heats incoming air
10	Valve 1	To control the water flow rate
11	Valve 2	To control the air flow rate
12	Valve 3	Drain valve

Commented [DL6]: Include the 6 thermocouples and the reader.

Commented [AZ7]: Steam valve is an important component since you will use it to increase the air temperature.

Commented [DL8]: Unit and the maximum capacity.

Commented [DL9]: Unit and the maximum capacity.

2.2 Materials and Supplies:

Table 2 : The important materials and supplies used in the experiment

#	The Material and Supplies	Description
1	Compressed Air	
2	Water	
3	Electronic hydrometer	Measures the temperature of the air exiting the humidification column
4	Thermocouple	Measures the temperature of incoming air and water, and the temperature of the water throughout the column

Commented [DL10]: 4.5/5
Include the ladder and the steam.

Commented [DL11]: Be specific in the apparatus section.

2.3 Experimental Plan:

> Before starting the experiment:

1. Measure the humidity of the incoming air using the wet and dry bulb thermometer or electronic hygrometer
2. Measure the temperature of the incoming and outgoing air

> The experiment:

1. Fill the tank with water
2. Turn on water flow at a sufficiently high flow rate to cover the entire inner surface of the column
3. Turn on the house supply in order to establish the unheated air flow at **3 SCFPM**
4. Allow the temperature of the water to reach the steady state
5. Record the water and air temperatures at the inlet, along the column and the air humidity at the outlet
6. Repeat the same steps using two different flow rates (**6 SCFPM and 9 SCFPM**)
7. Set the air flow to **6 SCFPM**
8. Now, turn on the steam in order to increase the temperature of the air flow
9. Repeat step 3 through 6 using the heated air flow
10. Turn off air and water flow
11. Clean up

Independent Variables:

Air flow rate, Liquid flow rate, temperature

Dependent Variables:

Humidity in air from top column, Temperature throw the column

Commented [DL12]: 9/10

Commented [AZ13]: Good

Commented [DL14]: How much?

Commented [AZ15]: House air supply

Commented [DL16]: What's the criteria of the steady state?

Commented [DL17]: You could obtain 2 different temperatures for air through different steam flow rates.

Commented [AZ18]: Drain the water using the pump and drain valve.

Project Safety Evaluation

Project Title: Humidification

Date: 01/21/2020

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: <u>pumps</u>
Potential mechanical hazards?	Yes	If yes, identify conditions: Fall hazard if it is needed to step on the <u>table</u> , <u>pumps</u>
Hazardous/toxic chemicals involved?	No	If yes, identify conditions:
<u>Incompatible chemicals involved?</u>	No	If yes, identify conditions:
Extra eye protection required?	No	If yes, specify splash goggles or face shield:
Gloves required?	No	If yes, specify type:
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: <u>CAS-7732-18-5</u> , <u>CAS-7782-44-7</u> , <u>CAS-7727-37-9</u>
Process Parameters	Max Expected	List location and precautions
Temperature (°C)	<u>100</u>	Steam. There are 7 thermocouples so temperature will vary
Pressure (psig)	Vary	<u>Pressure gauge goes up to 160 psig</u>

Commented [DL19]: 8.5/10

Commented [DL20]: Be specific.

Commented [AZ21]: Don't step on the table! Use the stepladder.

Commented [DL22]: Should be temperature.

Commented [AZ23]: Instead of using the separate CAS numbers for oxygen and nitrogen, use the one for compressed air: CAS 132259-10-0

Commented [DL24]: Maximum 120 degree Celsius.

Commented [DL25]: Only 40 psig house supply is used.

Commented [AZ26]: Maximum house supply is 120psig.

Safety Controls	yes/no	If yes, Provide description
	yes	Pressure gauges, thermocouple on cooling water, rotameter

Commented [AZ27]: Mention air shutoff valve, steam shutoff valve, pump switch

Commented [DL28]: The flow rate and temperature are controlled.
20-amp surge protector.
A regulator is attached to the system to allow complete control of air to prevent high pressure build up.

I have read relevant background material for the Unit Operations Laboratory entitled:
 “ Humidification ” and understand the hazards associated with conducting this
 experiment. I have planned out my experimental work in accordance to standards and acceptable
 safety practices and will conduct all of my experimental work in a careful and safe manner. I will
 also be aware of my surroundings, my group members, and other lab students, and will look out for
 their safety as well.

<u>Signature of Team member</u>	<u>Date</u>
Signature of Team member Ahmad Abusubaih	Date 01/21/2020
Signature of Team member Ahmed Alghamdi	Date 01/21/2020
Signature of Team member Fatimah AL Ibrahim	Date 01/21/2020
Signature of Team member Velia Palma	Date 01/21/2020
Signature of Team member Christian Sanchez	Date 01/21/2020